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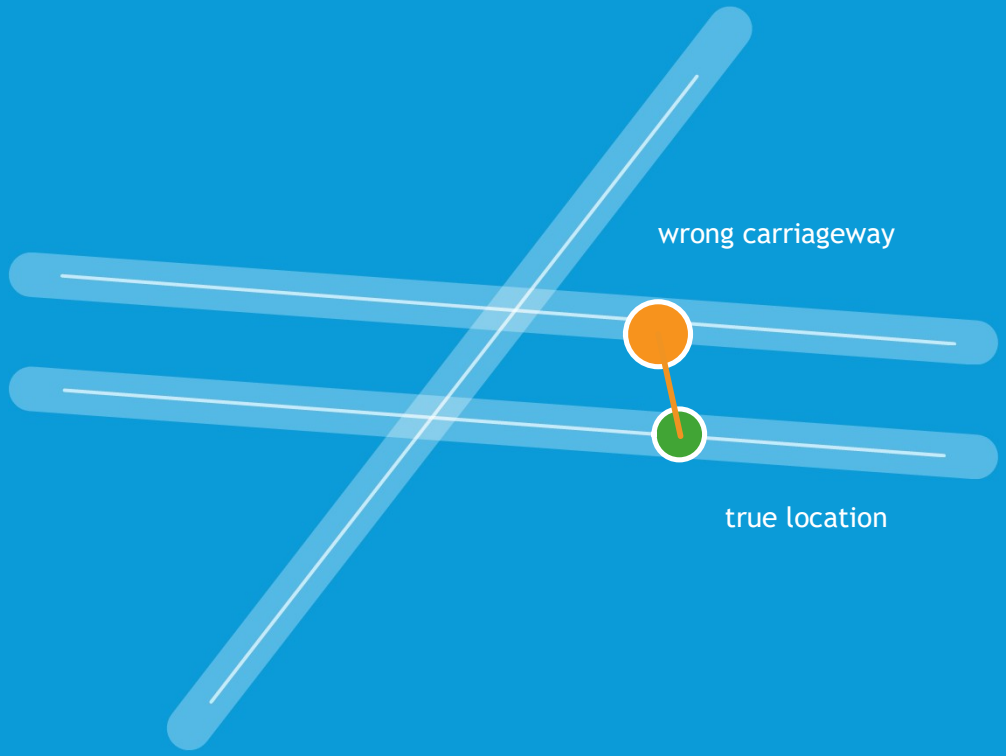
RTTI Data Quality Parameters

Abel Gabor · TISA · FOSS4G Belgium 2026

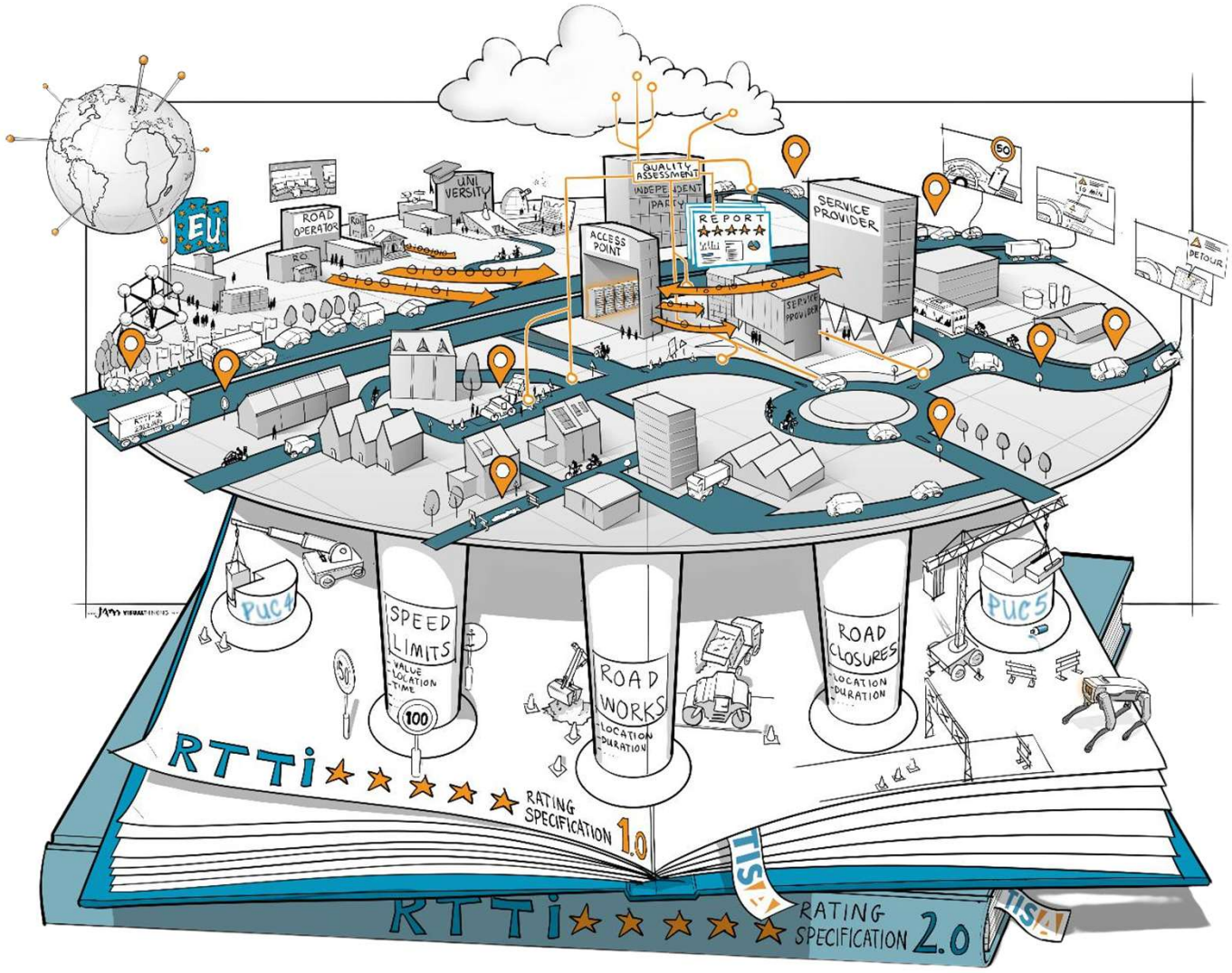
RTTI Data Quality Parameters

What they measure and why they matter

A traffic event can be close in coordinates but wrong in the road network.



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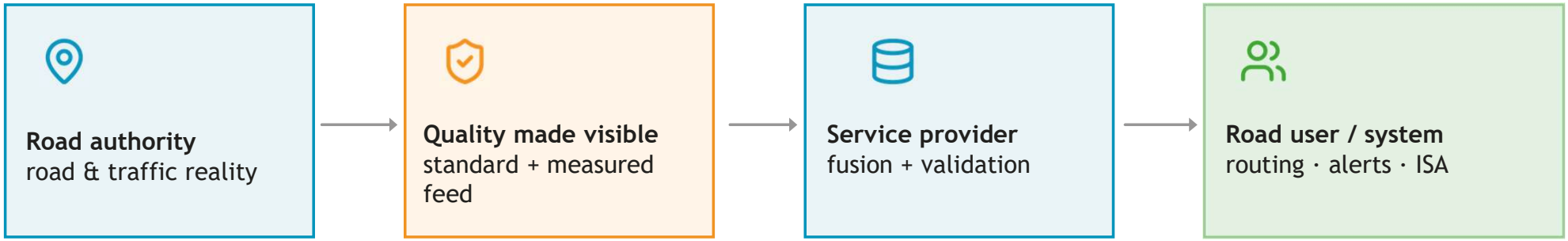
Available is not automatically usable

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Open access answers: “Can I obtain the feed?”

Data quality answers: “Can I automate its reuse?”

A shared quality language makes the road-data handoff inspectable.



The RTTI 5-Star Rating assesses the data handoff — not the finished navigation experience.

Source: TISA RTTI 5-Star Rating Specification, Vol. 1 — Scope and Applicability

Available is not automatically usable

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Three road-data foundations. One transparent improvement path.

01

Static speed limits

Trusted road attributes for maps and safety functions.

02

Roadworks

Planned and unplanned events that affect traffic flow.

03

Full road closures

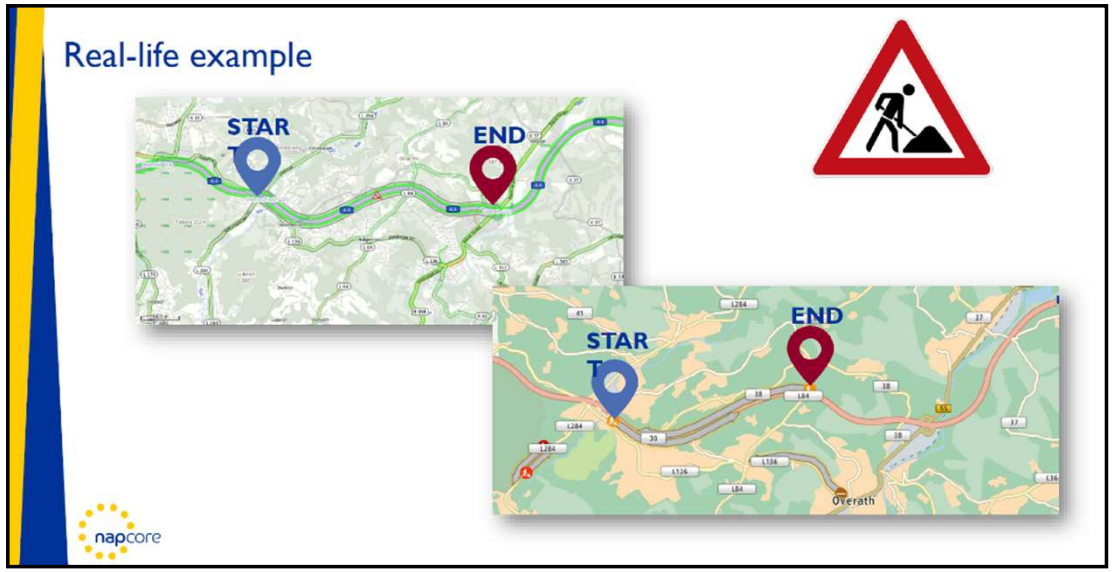
Planned and unplanned events requiring alternative routing.

Measurable quality requirements + interoperable exchange

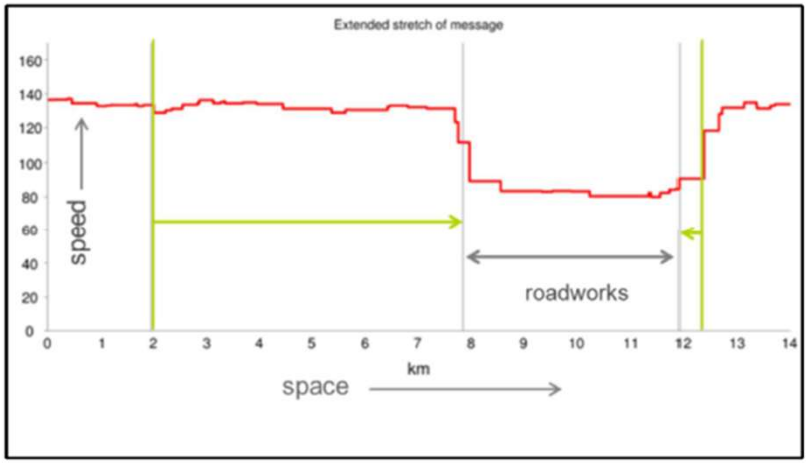
A voluntary framework for data quality — not a consumer score and not a rating of the final service.

Source: TISA RTTI 5-Star Rating Specification, Vol. 1 — Foreword and Priority Use Cases

SP Quality Assurance roadworks extent



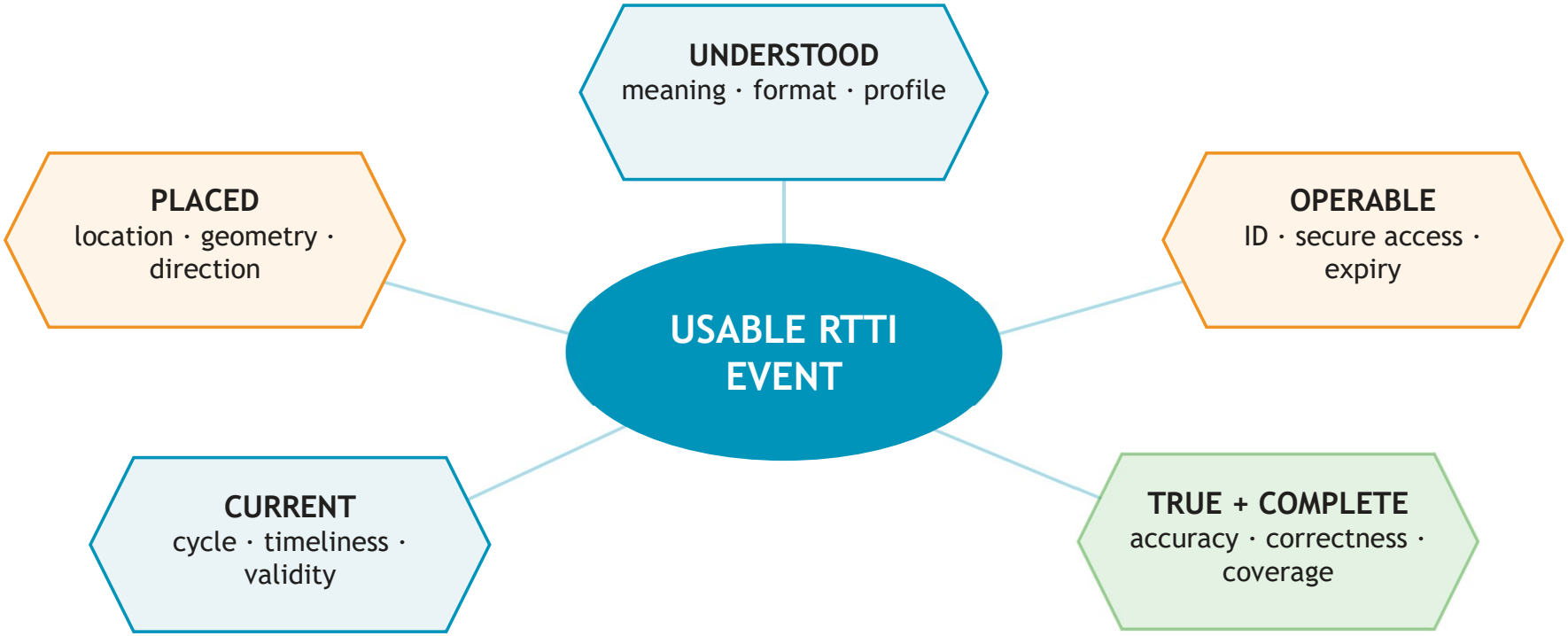
Slide of NAPCORE presentation Peter Lubrich, BASt @ TISA September 2023



Source: TomTom, ATEC ITS France Congress, 2014, TomTom Brouwer paper

Quality is a system, not a single number

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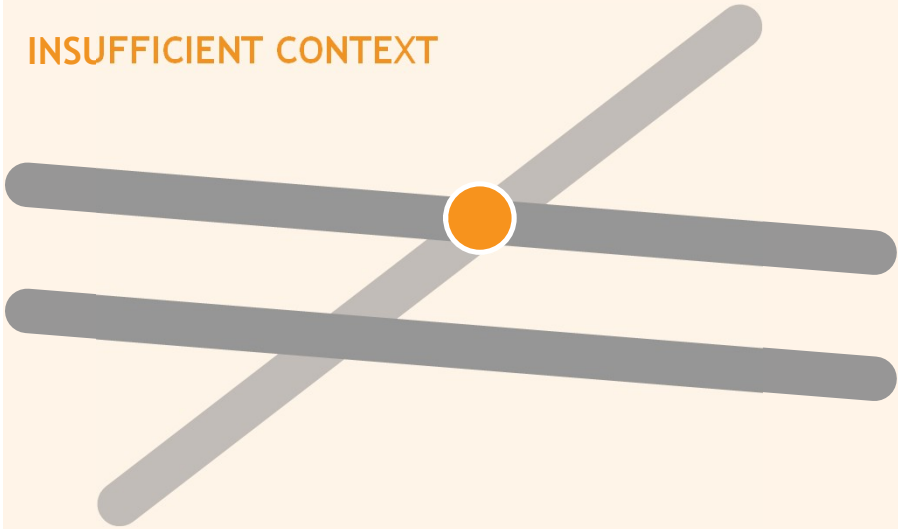
One weak critical dimension can block reliable automated use.

Source: TISA RTTI 5-Star Rating Specification, Vol. 1 — Criteria and Annex 2

Can the receiving system put it on the right road?

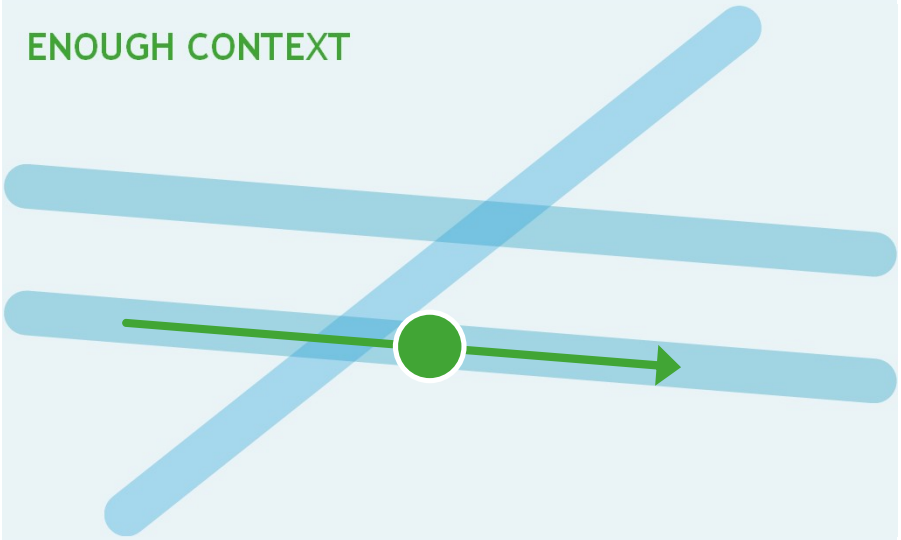
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INSUFFICIENT CONTEXT



Point: 50.8500, 4.3500
“Roadworks on A2”
Which carriageway? Which direction? What extent?

ENOUGH CONTEXT



Linear location + direction
start / end / via points · lane context
Map-independent reference: OpenLR

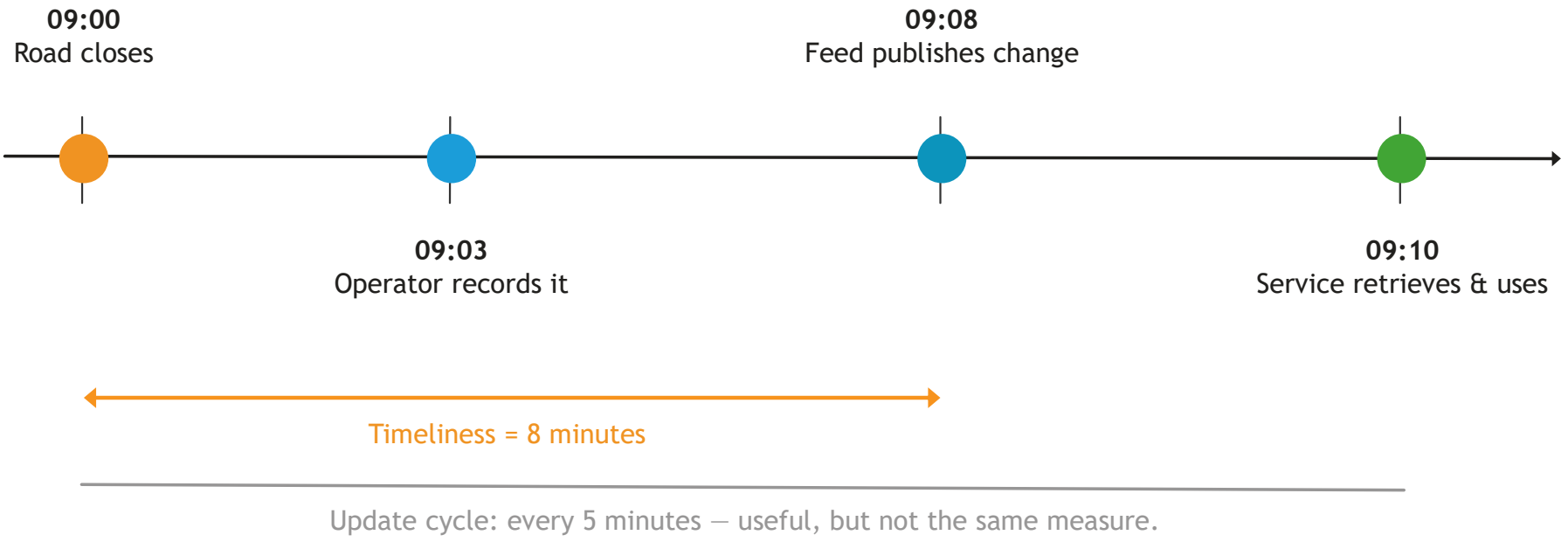
Spatial quality means correct network placement — not merely a coordinate inside a tolerance

Source: TISA RTTI 5-Star Rating Specification, Vol. 1 — Spatial Accuracy; Vol. 3 — Annex 1

Is it true at the time it reaches the driver?

Update cycle, timeliness and validity measure different things.

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A frequent polling cycle cannot recover a late source update. Validity says when the event actually applies.

Source: TISA RTTI 5-Star Rating Specification, Vol. 1 — Accuracy in Time and Update Cycle vs Timeliness

Is the feed dependable enough to automate?

Four questions separate a plausible record from a reliable one.

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Accuracy

How close is the reported location or time to reality?

Correctness

Are the event, direction, impact and timing actually true?

Completeness

Are expected coverage and required attributes present?

Lifecycle

Can the same event be updated and removed reliably?

BAD LIFECYCLE

Closure ID changes → receiver sees two closures

GOOD LIFECYCLE

Stable ID + update + removal → one current state

Source: TISA RTTI 5-Star Rating Specification, Vol. 1 — Terminology and Annex 2

One roadworks event: from ambiguous record to usable data

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MINIMAL / AMBIGUOUS

Roadworks on A2
Location: 50.8500, 4.3500
Start: 1 Oct
End: 31 Oct

Missing:
- extent
- direction
- schedule
- lane impact
- stable ID

USABLE / CONTEXT-RICH

ID: RW-2026-1042 (stable)
Type: planned roadworks
Location: start → via points → end
Direction: eastbound
Impact: right lane closed; residual width 2.5 m
Validity: 1–31 Oct, Mon–Fri 22:00–06:00
Published / updated: timestamp

Fictional pseudorecord - illustrative, not an actual authority feed.

A usable event record answers: what, where, which direction, when, what impact — and whether this is the same event.

Source: TISA RTTI 5-Star Rating Specification, Vol. 3 — Roadworks requirements and Annex 1

What the stars mean: measurable progress, not decoration

A weighted assessment makes improvement visible without treating every parameter as equal.

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Critical parameters carry more weight. A 2.9 is not rounded up to 3.

Source: TISA RTTI 5-Star Rating Specification, Vol. 1 — Rating Levels, Structured Scoring and Rounding

Open standards turn quality into reusable infrastructure

The value is not the acronym. The value is fewer one-off interpretations.

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Dynamic events

DATEX II

Roadworks · full road closures
Structured events, updates and impacts

Shared semantics + consistent profiles

Static attributes

TN-ITS

Static speed limits
Authoritative road-attribute updates

Faster integration · fewer bespoke parsers · less ambiguity in downstream processing

Source: TISA RTTI 5-Star Rating Specification, Vol. 1 — Approved Data Formats and Protocols for Data Exchange

A pragmatic improvement path for data publishers

Start where reuse fails. Improve iteratively; do not wait for perfect data.

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01 Inventory
What exists, who owns it, and how is it published?

02 Measure
Test spatial, temporal and lifecycle quality.

03 Prioritise
Fix the gaps that block automated reuse.

04 Expose & improve
Share quality evidence and improve the next constraint.

PRIORITISE THE OPERATIONAL BLOCKERS

geometry + direction · event time + timeliness · IDs + expiry · standard / profile use

Source: TISA RTTI 5-Star Rating Specification, Vol. 1 — Benefits, Data Segmentation and Outlook

Three takeaways

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01

Quality is more than a coordinate or an API endpoint.

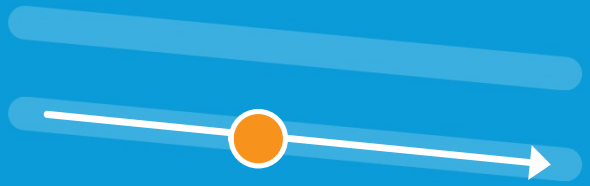
02

Spatial, temporal and lifecycle context determines automated reuse.

03

A transparent rating creates a shared path from publication to use.

Let’s make road data easier to trust and reuse.



Advancing Data Quality Together

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Vision

A world where traffic and travel information is 100% correct, 100% of the time on all roads.

Our Mission & Purpose

MISSION

To lead the development of trusted traffic & travel standards and harmonised services for our members.

PURPOSE

Ensure travel is safe, efficient and sustainable for everyone, everywhere.

Our Global Membership

ITS Service and Solution Providers, User Device Manufacturers, Public Sector Stakeholders and Vehicle Manufacturers.



RTTI 5-Star Rating Specification



Functional Road Classifications - Introduction

Online calculation tool shared

Thank you!



Ábel Gábor

Specification/Standards Editor

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a.gabor@tisa.org

www.tisa.org

Appendix A: Illustrative thresholds

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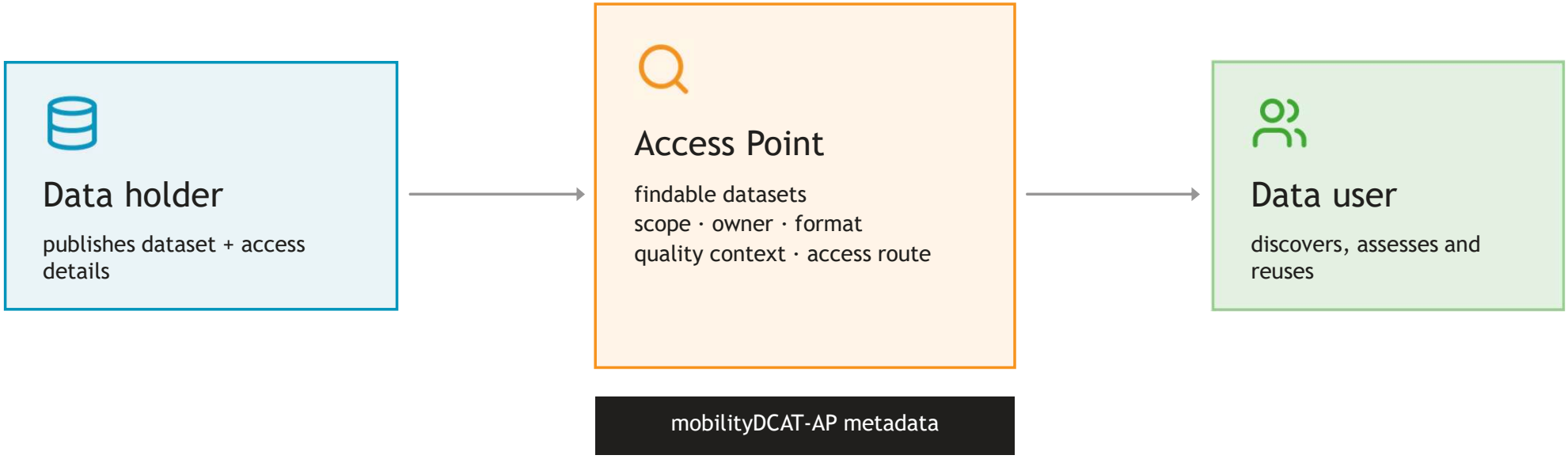
Selected examples only. Overall ratings use multiple weighted criteria; exact assessment methodology remains separate.

Use case	Illustrative 3-star level	Illustrative 5-star level	Why it matters
Static speed limits	Weekly update / timeliness; at least 90% correctness and completeness; within 10 m location accuracy	Daily update / timeliness; at least 99% correctness and completeness; within 1 m accuracy	Digital road attributes become more reliable for maps and safety functions.
Planned roadworks FRC1–4	Daily; max 24h timeliness; within 250 m; at least 80% correctness and completeness	Hourly; max 1h; within 50 m; at least 90% correctness and completeness	Location and timing must support meaningful routing decisions.
Unplanned events	Max 10 min update / timeliness	Max 1 min update / timeliness	Dynamic disruptions lose operational value rapidly when delayed.

These examples explain the scale of progress; they are not a substitute for the full per-use-case requirements.

Source: TISA RTTI 5-Star Rating Specification, Vols. 2–4 — requirement tables

Appendix B: Data discovery and metadata



Discoverability says where and how to obtain data. Quality evidence tells whether automated reuse is appropriate.

Source: TISA RTTI 5-Star Rating Specification, Vol. 1 — Data Delivery, Formats and Protocols